COURSEWORK ASSIGNMENT SUBMISSION

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Subject: Software Engineering & Project Management **Day:** Day 21 — Project Management Fundamentals (WBS & Scheduling)

Section 11: Classroom Activity — Plan CEMS

Comprehensive project initiation and Work Breakdown Structure (WBS) plan for the College Event Management System.

1. Project Charter & Measurable Objectives

- **Project Objective:** Deliver a centralized, responsive web platform for campus event management within a 4-week delivery timeframe.
- **Success Criterion 1:** Support 100% of campus technical festival event registrations with zero duplicate bookings.
- **Success Criterion 2:** Achieve sub-1.5 second page load times under 300 concurrent student users during registration opening.
- **Success Criterion 3:** Zero critical or high-severity defects remaining open at the formal handover review.

2. Three-Level Work Breakdown Structure (WBS)

Level 1 (Phase)	Level 2 (Deliverable)	Level 3 (Work Package ID & Scope)
1.0 Requirements	1.1 Stakeholder Analysis	WP-1.1.1 Conduct 10 coordinator interviews; WP-1.1.2 Student survey.
1.0 Requirements	1.2 Specification	WP-1.2.1 Draft formal SRS document; WP-1.2.2 CCB Baseline approval.
2.0 System Design	2.1 Architecture	WP-2.1.1 3-Tier Layered Architecture ADR; WP-2.1.2 API contract design.
2.0 System Design	2.2 Database Design	WP-2.2.1 Entity Relationship Diagram; WP-2.2.2 PostgreSQL schema DDL.
3.0 Implementation	3.1 Core Backend API	WP-3.1.1 Auth & RBAC service; WP-3.1.2 Booking engine with row locks.
3.0 Implementation	3.2 Web Frontend	WP-3.2.1 Responsive event catalog; WP-3.2.2 QR Ticket modal & dashboard.
4.0 Quality & Delivery	4.1 Testing & QA	WP-4.1.1 Unit & integration test suites; WP-4.1.2 JMeter load testing.
4.0 Quality & Delivery	4.2 Deployment	WP-4.2.1 Docker containerization; WP-4.2.2 Production release sign-off.

3. Activity Dependency & Four-Week Gantt Schedule

Activity Sequencing: A: Requirements (Days 1–5) -> B: System Design (Days 6–10, Pred: A) -> C: Backend Dev (Days 11–18, Pred: B) -> D: Frontend Dev (Days 13–20, Pred: B) -> E: Integration & Testing (Days 21–25, Pred: C, D) -> F: User Acceptance & Deployment (Days 26–28, Pred: E).
Major Milestones: M1: SRS Sign-off (End of Week 1); M2: Alpha Integration Freeze (End of Week 3); M3: Production Go-Live (Day 28).

Section 14: Homework — Project Charter & WBS for Library System

Project initiation charter, 12-element WBS, and risk response matrix for an automated Library Management System.

- **1. System Charter:** Automate catalog discovery, circulation loan desks, and fine management for a 50,000-volume university library. Target Go-Live: 30 working days.
- **2. 12-Package WBS Breakdown:** 1.1 Patron Management Module; 1.2 Catalog ISBN Search; 1.3 Barcode Scan Integration; 2.1 Loan Checkout Service; 2.2 Book Return & Renewal Engine; 2.3 Automated Overdue Fine Calculator; 3.1 Relational Schema & Indexing; 3.2 RFID Reader API Driver; 3.3 Student Portal UI; 4.1 Automated Circulation Unit Tests; 4.2 End-to-End Stress Test; 4.3 Production Server Migration.
- **3. Three Schedule Risks & Planned Responses:** (a) Delays in physical barcode scanner hardware delivery -> Response: Develop hardware abstraction mock interface to proceed with software development; (b) Scope creep from academic faculty demanding custom loan rules -> Response: Enforce strict Change Control Board review and defer custom rules to Release 2.0; (c) Database indexing bottlenecks under catalog search load -> Response: Provision staging Redis cache spike in Week 2.